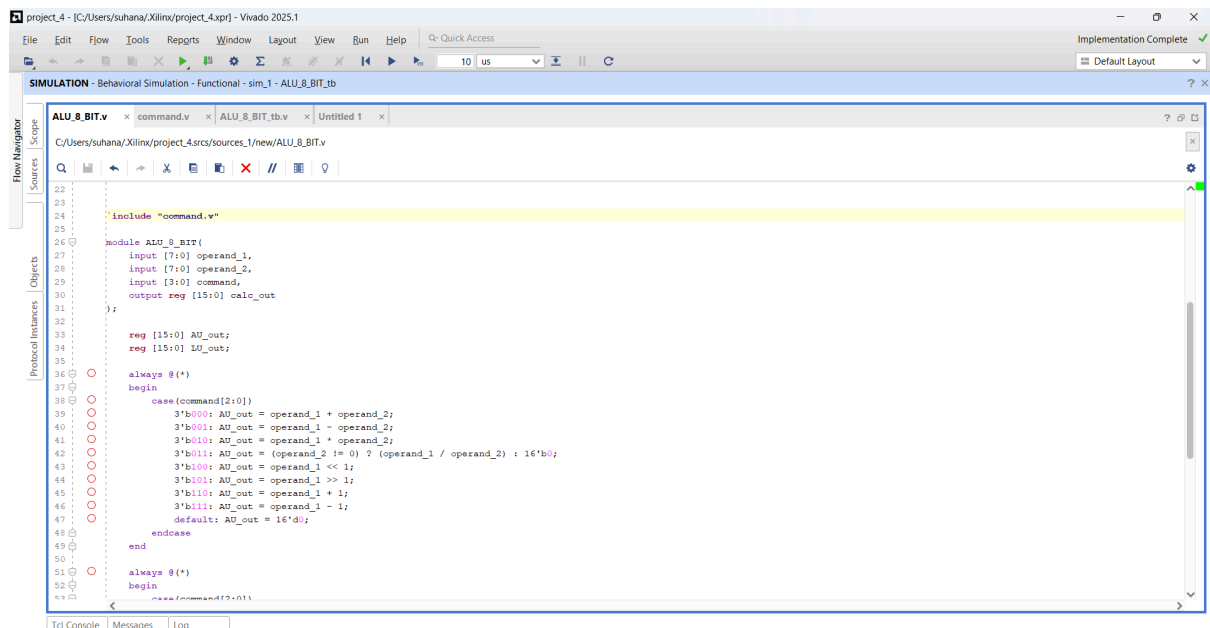
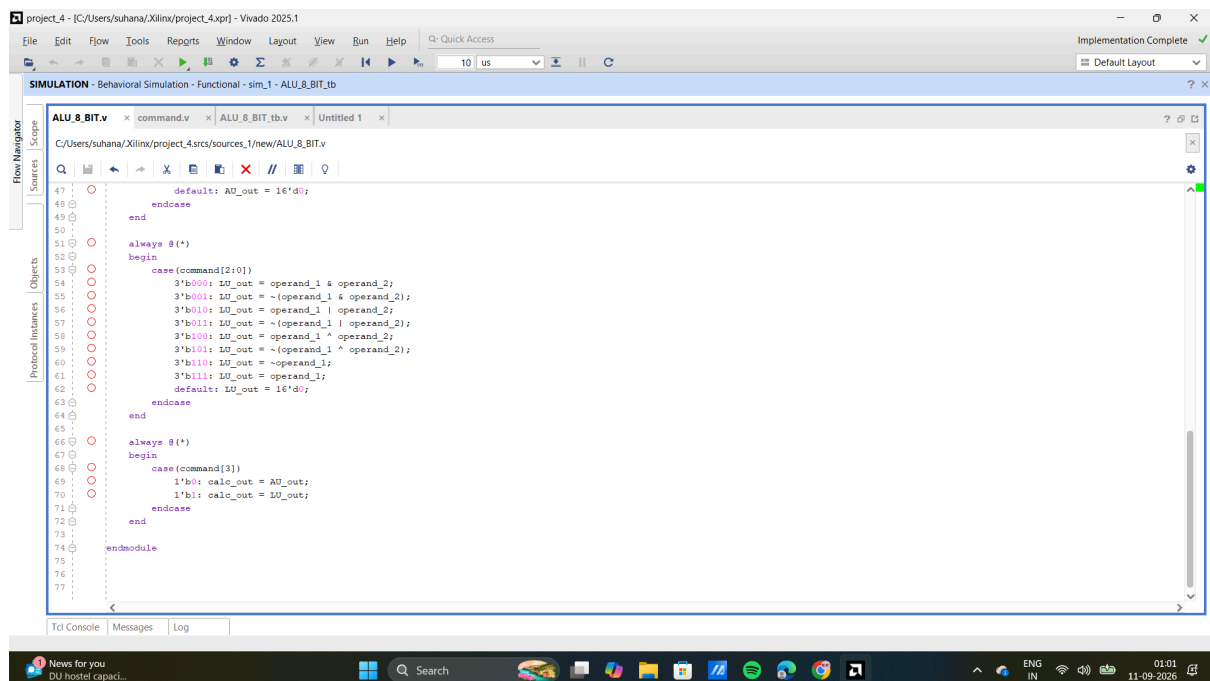


8 bit ALU in Verilog Xilinx Vivado

1. ALU_8_BIT.v

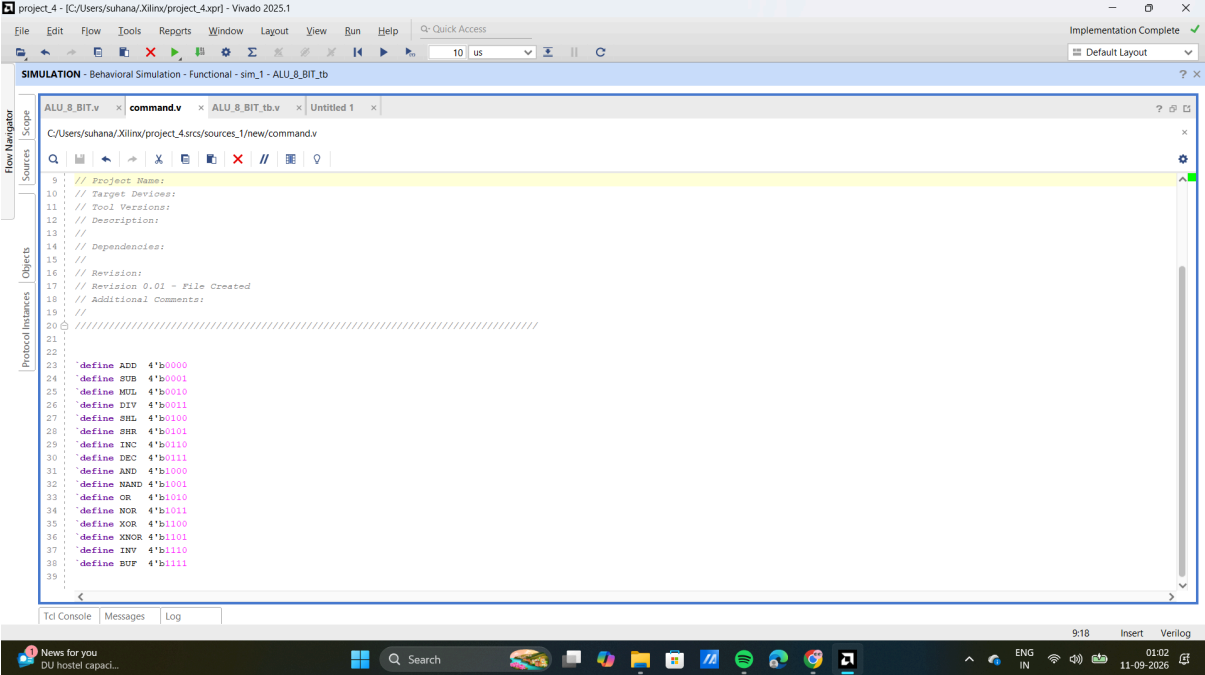


```
22
23
24 include "command.v"
25
26 module ALU_8_BIT(
27     input [7:0] operand_1,
28     input [7:0] operand_2,
29     input [3:0] command,
30     output reg [15:0] calc_out
31 );
32
33     reg [15:0] AU_out;
34     reg [15:0] LU_out;
35
36     always @(*)
37     begin
38         case(command[2:0])
39             3'b000: AU_out = operand_1 + operand_2;
40             3'b001: AU_out = operand_1 - operand_2;
41             3'b010: AU_out = operand_1 * operand_2;
42             3'b011: AU_out = (operand_2 != 0) ? (operand_1 / operand_2) : 16'b0;
43             3'b100: AU_out = operand_1 << 1;
44             3'b101: AU_out = operand_1 >> 1;
45             3'b110: AU_out = operand_1 + 1;
46             3'b111: AU_out = operand_1 - 1;
47             default: AU_out = 16'd0;
48         endcase
49     end
50
51     always @(*)
52     begin
53         // ... (code for LU_out calculation) ...
54     end
55 endmodule
```



```
47     default: AU_out = 16'd0;
48 endcase
49 end
50
51     always @(*)
52     begin
53         case(command[2:0])
54             3'b000: LU_out = operand_1 & operand_2;
55             3'b001: LU_out = ~(operand_1 & operand_2);
56             3'b010: LU_out = operand_1 | operand_2;
57             3'b011: LU_out = ~(operand_1 | operand_2);
58             3'b100: LU_out = operand_1 ^ operand_2;
59             3'b101: LU_out = ~(operand_1 ^ operand_2);
60             3'b110: LU_out = operand_1;
61             3'b111: LU_out = operand_2;
62             default: LU_out = 16'd0;
63         endcase
64     end
65
66     always @(*)
67     begin
68         case(command[3])
69             1'b1: calc_out = AU_out;
70             1'b0: calc_out = LU_out;
71         endcase
72     end
73 endmodule
74
75
76
77
```

2. command.v



The screenshot shows the Vivado IDE interface. The top menu bar includes File, Edit, Flow, Tools, Reports, Window, Layout, View, Run, Help, and a Quick Access search bar. The title bar indicates the project is 'project_4' located at 'C:/Users/suhana/Xilinx/project_4.xpr' in Vivado 2025.1. The main window is titled 'SIMULATION - Behavioral Simulation - Functional - sim_1 - ALU_8_BIT_tb'. The left sidebar contains a 'Flow Navigator' with tabs for 'Scope', 'Sources', 'Objects', and 'Protocol Instances'. The 'Sources' tab is active, showing a list of files: 'ALU_8_BIT.v', 'command.v', 'ALU_8_BIT_tb.v', and 'Untitled 1'. The 'command.v' file is selected and its content is displayed in the main editor. The code is a Verilog module with various logic definitions. The bottom status bar shows 'Tcl Console', 'Messages', and 'Log' tabs, along with system information like '9:18', 'Insert', 'Verilog', and the date '11-09-2026'.

```
9 // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 `define ADD 4'b0000
24 `define SUB 4'b0001
25 `define MUL 4'b0010
26 `define DIV 4'b0011
27 `define SHL 4'b0100
28 `define SHR 4'b0101
29 `define INC 4'b0110
30 `define DEC 4'b0111
31 `define AND 4'b1000
32 `define NAND 4'b1001
33 `define OR 4'b1010
34 `define NOR 4'b1011
35 `define XOR 4'b1100
36 `define XNOR 4'b1101
37 `define INV 4'b1110
38 `define BUF 4'b1111
39
```

3. ALU_8_BIT.tb

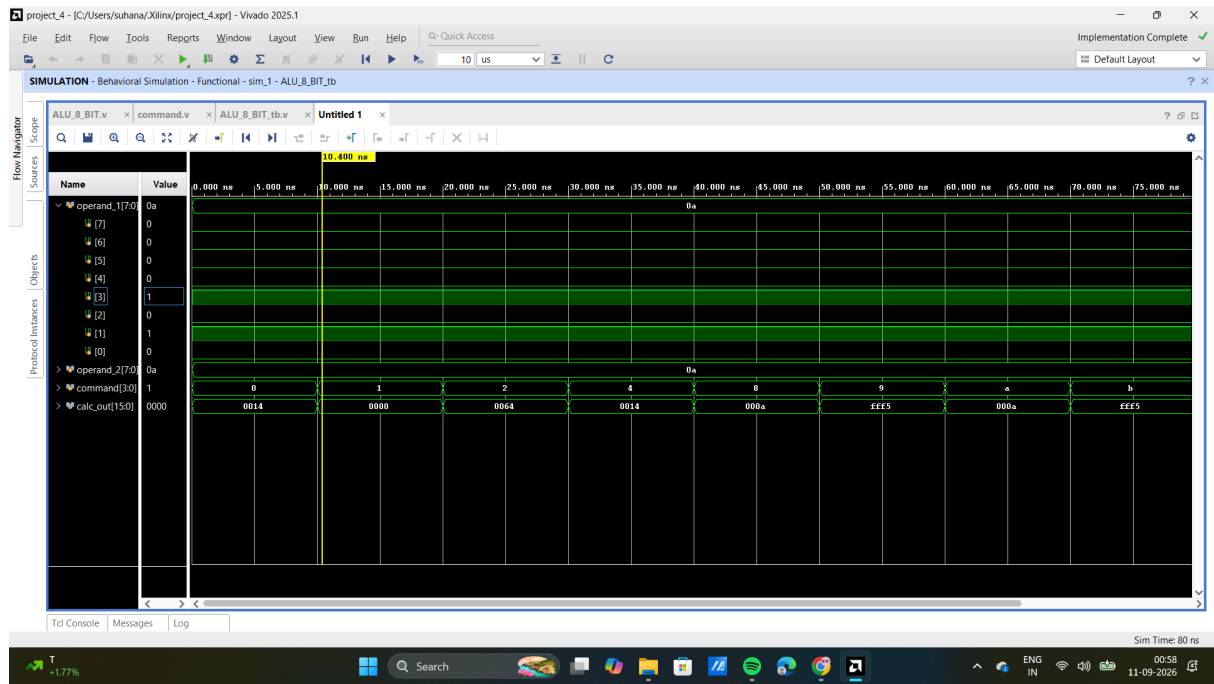
```
23 module ALU_8_BIT_tb;
24
25     reg [7:0] operand_1, operand_2;
26     reg [3:0] command;
27     wire [15:0] calc_out;
28
29     ALU_8_BIT DUT(
30         .operand_1(operand_1),
31         .operand_2(operand_2),
32         .command(command),
33         .calc_out(calc_out)
34     );
35
36     initial begin
37         operand_1 = 8'd10;
38         operand_2 = 8'd10;
39
40         command = 4'b0000; #10;
41         command = 4'b0001; #10;
42         command = 4'b0010; #10;
43         command = 4'b0100; #10;
44         command = 4'b1000; #10;
45         command = 4'b1001; #10;
46         command = 4'b1010; #10;
47         command = 4'b1011; #10;
48
49         $finish;
50     end
51
52 endmodule
53
```

4. Output waveform of tb 1

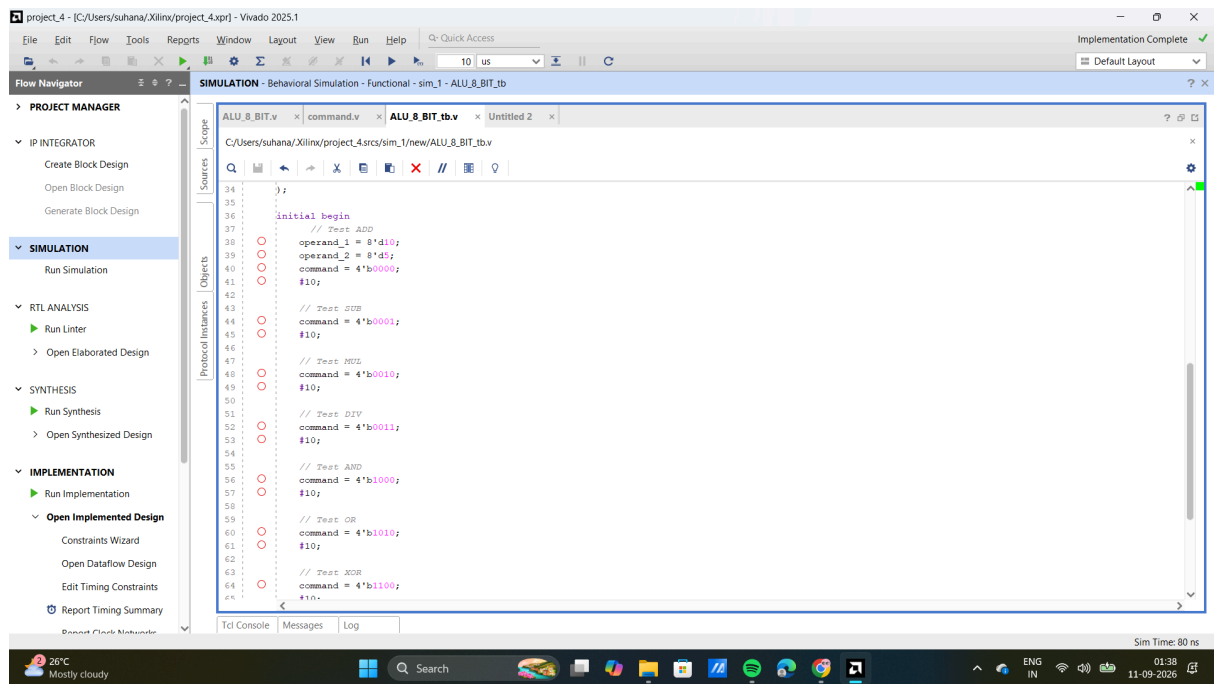
3. Your arithmetic commands

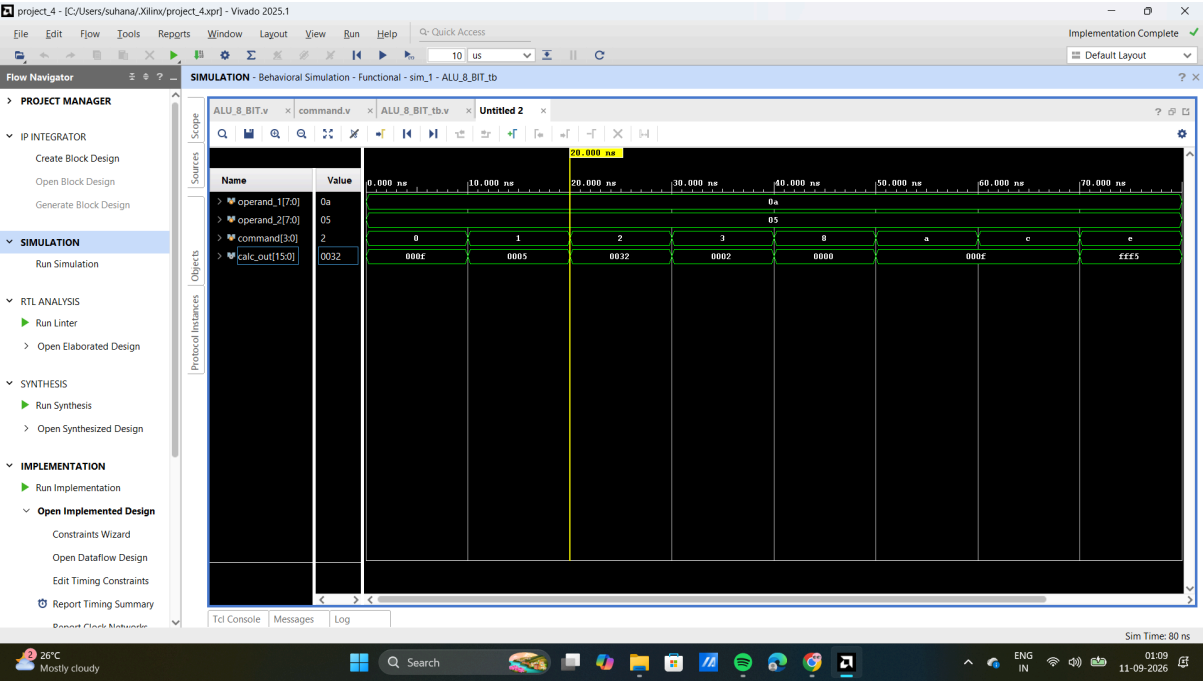
From the ALU code you showed:

Command	Operation	With 10,10	Output
0000	ADD	10 + 10	20 = 0014
0001	SUB	10 - 10	0 = 0000
0010	MUL	10 × 10	100 = 0064
0011	DIV	10 / 10	1 = 0001
0100	Left shift	10 << 1	20 = 0014
0101	Right shift	10 >> 1	5 = 0005
0110	+1	10 + 1	11 = 000B
0111	-1	10 - 1	9 = 0009



5. Tb 2 code:





6. TB code 3

The image displays two screenshots of the Vivado IDE, showing the implementation and simulation of a testbench for an 8-bit ALU.

Top Screenshot: Testbench Code

The top screenshot shows the testbench code for `ALU_8_BIT_tb.v`. The code is a Verilog testbench that initializes two 8-bit operands and an 8-bit command, then performs a series of operations (AND, NAND, OR, NOR, XOR, XNOR, and NOT) on the operands using the command to produce the output `calc_out`.

```
initial begin
    // Give inputs
    operand_1 = 8'd12; // 00001100
    operand_2 = 8'd5;  // 00000101

    // AND
    command = 4'b1000;
    #10;

    // NAND
    command = 4'b1001;
    #10;

    // OR
    command = 4'b1010;
    #10;

    // NOR
    command = 4'b1011;
    #10;

    // XOR
    command = 4'b1100;
    #10;

    // XNOR
    command = 4'b1101;
    #10;

    // NOT operand_1
    command = 4'b1110;
    #10;

    // operand_1
```

Bottom Screenshot: Simulation Waveform

The bottom screenshot shows the simulation waveform for the testbench. The waveform displays the values of the inputs (`operand_1`, `operand_2`, `command`) and the output (`calc_out`) over time. The time scale is in nanoseconds (ns), ranging from 0.000 ns to 80.000 ns. The output `calc_out` is shown in hexadecimal (hex) and binary (bin) formats.

Time (ns)	operand_1 [7:0]	operand_2 [7:0]	command [3:0]	calc_out [15:0]
0.000	0c	05	8	0004
10.000	0c	05	9	eeeb
20.000	0c	05	a	0004
30.000	0c	05	b	eeeb
40.000	0c	05	c	0009
50.000	0c	05	d	eeeb
60.000	0c	05	e	eeeb
70.000	0c	05	f	000c

7. schematic

